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Disaster evacuation of the old city of Nicosia ☆

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ABSTRACT

Historic urban centers face unique evacuation challenges due to dense morphology, limited network redundancy, and cultural heritage constraints that restrict infrastructural modification. This study presents an exploratory, decision-support analysis of evacuation dynamics in the walled Old Town of Nicosia, Cyprus, using a high-fidelity digital twin and a hybrid agent-based simulation framework integrating pedestrian and vehicular movement. Rather than modeling a specific hazard, the study evaluates how structural stressors commonly produced by disasters, including exit closures, road blockages, and shelter saturation, influence evacuation clearance, congestion formation, and overall system robustness. Controlled simulation experiments examine the sensitivity of evacuation performance to network disruptions, the effectiveness of existing and expanded shelter configurations, and the role of population familiarity and guidance in constrained historic environments. Results show that evacuation outcomes are highly sensitive to a small number of critical exits and that uncoordinated increases in shelter capacity, without adaptive guidance, may increase congestion and clearance times. Findings also reveal nonlinear congestion dynamics and long-tailed evacuation delays that persist under idealized conditions. The study demonstrates the value of high-resolution digital twins as exploratory tools for identifying structural vulnerabilities, evaluating mitigation strategies, and supporting risk-informed evacuation planning in historic districts exposed to multiple hazards.

1. Introduction

Historic urban centers embody a persistent tension between cultural heritage preservation and urban vulnerability. Dense morphology, limited network redundancy, and regulatory constraints on structural modification often restrict the range of feasible risk-reduction interventions. As a result, evacuation planning in such environments must operate within spatial and institutional limitations that differ fundamentally from those of modern cities.

This tension is particularly evident in the walled Old Town of Nicosia, Cyprus. Enclosed by 16th-century Venetian fortifications and partially bordered by a militarized buffer zone, the historic core is accessible through only six operational gates. Its irregular medieval street network, narrow corridors, pedestrian-only segments, and one-way constraints create asymmetric flow patterns and limited alternative routing options. While these features contribute to the city's heritage value, they also introduce structural bottlenecks that can severely constrain evacuation under emergency conditions.

The urgency of addressing these vulnerabilities has increased in the Eastern Mediterranean, where climate-related and human-induced hazards are intensifying. Although Nicosia participates in the UNDRR Making Cities Resilient 2030 initiative, existing emergency planning frameworks remain largely schematic and do not incorporate high-resolution, data-driven evacuation modeling at city scale. Previous research on evacuation in Mediterranean heritage contexts often focuses on single hazards, simplified networks, or purely pedestrian dynamics, frequently overlooking political boundaries and mixed pedestrian–vehicle interactions [1]. Consequently, a gap persists in understanding how historic morphology, limited exit redundancy, and real-time decision-making jointly influence evacuation performance in constrained heritage environments.

More broadly, while urban evacuation modeling has advanced substantially in recent years [2,3], no systematic, city-scale simulation has been conducted for the walled Old Town of Nicosia. The combination of fortification-imposed exits, one-way circulation, a militarized zone on the northern boundary (see Fig. 1), and high seasonal population

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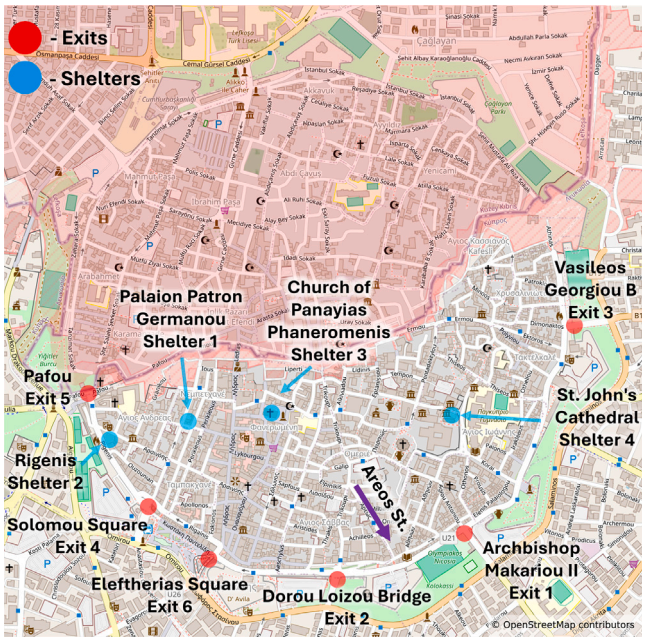


Fig. 1. Geo-referenced 3D map of Nicosia's Old Town, showing city gates (exits), active shelters, and the military buffer zone to the north.

variability makes this context a representative yet underexplored case for studying evacuation dynamics in historic cities.

1.1. Multi-hazard evacuation challenges in historic cities

Historic districts are rarely exposed to a single hazard type, and risk often emerges from interacting or cascading events [1,4]. Earthquakes may generate debris-induced road blockages and façade collapses, while floods introduce depth-dependent walkability constraints and progressive route loss [5]. In morphologically constrained historic environments, such hazard-specific effects interact with limited network redundancy, potentially producing nonlinear congestion and long-tailed evacuation delays.

Although the present study does not simulate a specific hazard process, it focuses on structural stressors that commonly arise across hazard types: gate closures, road blockages, and shelter saturation. By abstracting from hazard triggers while retaining their systemic consequences, the analysis provides hazard-agnostic insights into evacuation vulnerability in historic environments. The framework is intentionally extensible, enabling the integration of debris fields, flood extents, or fire spread models in future work.

Different hazards can create different evacuation conditions in dense historic environments. Earthquakes can cause sudden infrastructure damage and debris that block narrow streets. Floods gradually reduce road accessibility as water levels rise. Urban fires, in contrast, require rapid evacuation under strong time pressure and may lead to more chaotic and panic-driven movement patterns. Previous studies in historic city centers such as Florence (Italy) [1], Dubrovnik (Croatia) under seismic risk [6], and Beirut (Lebanon) in earthquake evacuation simulations [7], as well as other Mediterranean heritage contexts [5], highlight how warning time, infrastructure disruption, and population behavioral responses vary, shaping evacuation dynamics and congestion formation.

1.2. Research questions

This study addresses the following research questions:

RQ1: How do limited exit availability and constrained street morphology influence evacuation clearance times and congestion patterns in a historic walled city?

RQ2: How do shelter capacity and spatial distribution affect evacuation performance under conditions of network stress?

RQ3: To what extent does population familiarity and guidance modify evacuation outcomes in dense heritage environments?

Rather than predicting evacuation outcomes for a specific disaster scenario, the objective is to identify structural vulnerabilities and decision-relevant sensitivities that can inform risk-informed planning.

1.3. Study approach and contributions

To investigate these questions, we employ a high-fidelity, geo-referenced 3D digital twin of Nicosia's Old Town.[1] The simulation framework integrates detailed spatial data with a hybrid agent-based model capable of representing both pedestrian and vehicular movement within the same constrained environment. Pedestrian and vehicle agents operate on layered navigation meshes that encode one-way restrictions, pedestrian-only segments, and exit constraints. Behavioral parameters are calibrated using available traffic counts and density observations to approximate realistic movement dynamics.

A series of controlled simulation experiments was conducted under generalized stress conditions, including selective gate closures, road blockages, and alternative shelter configurations. These scenarios isolate the effects of exit redundancy, network asymmetry, and information availability on clearance performance and congestion formation. By integrating physically realistic agent behavior with accurate topographical data, we produced actionable insights for urban planners, emergency services, and policy-makers. Our simulation experiments reveal that evacuating Nicosia's semicircular walled core can become significantly delayed under certain conditions, and that adding infrastructure, without first providing clear guidance to residents, fails to improve evacuation outcomes. The resulting simulation dataset offers a valuable foundation for future academic research in crowd dynamics and evacuation modeling. The contributions of this study are threefold:

- Development of a high-resolution digital twin of Nicosia's Old Town that integrates geo-referenced road network data with a hybrid pedestrian–vehicle agent-based simulation framework.
- Systematic evaluation of evacuation performance under controlled stress conditions, quantifying clearance times, congestion hotspots, and shelter saturation effects.
- Identification of structural sensitivities in historic evacuation topology, demonstrating that performance is dominated by a small number of critical exits and strongly mediated by guidance and information availability.

By positioning evacuation as a systemic interaction between urban morphology, infrastructure constraints, and behavioral adaptation, this work advances the use of digital twins as exploratory decision-support tools for disaster risk reduction in heritage cities.

2. Literature review and background

Evacuation planning in dense, historic urban environments presents distinct challenges that transcend conventional transportation and civil engineering paradigms. In particular, the irregular morphology, restricted egress points, and mixed traffic patterns characteristic of heritage city centers demand a multidisciplinary approach.

[1] https://inicosia.cyens.org.cy/

In general, agent-based frameworks such as those presented in [8] and [9] have significantly advanced the realism of crowd movement by integrating psychological and perceptual models. These systems support dynamic obstacle avoidance, group behavior, and real-time replanning, key capabilities when simulating evacuation through constrained, evolving spaces. Narain et al. [10] and Karamouzas et al. [11] introduced data-driven and velocity-based crowd models, respectively, enabling smoother, more predictive simulations that scale efficiently to dense urban populations.

In terms of city evacuations scenarios, early contributions to pedestrian dynamics include the work of Helbing et al. [12], who introduced the Social Force Model (SFM), a continuous micro-simulation framework that has since been widely adapted for emergency scenarios. Extensions of this model by Yang et al. [3] introduced behavioral heuristics, including panic escalation and adaptive exit selection, to better reflect decision-making in disaster contexts. Bernardini et al. [5] further coupled pedestrian exit models with flood simulations, demonstrating that historical morphologies can produce localized deviations in evacuation times exceeding 15%. Finally, works like [13] presents an agent-based simulation model that incorporates the concept of affordance to analyze pedestrian evacuation behavior during emergencies.

Multi-modal evacuation, where both pedestrian and vehicular flows must be considered, introduces additional complexity. Zhang and Chang [14] proposed a dynamic network flow model that jointly optimizes evacuation routes while resolving pedestrian–vehicle conflicts at intersections. Ronchi and Nilsson [15] integrated vehicle dynamics into the FDS+Evac environment, revealing that spontaneous mode-switching behaviors (e.g., abandoning vehicles mid-evacuation) can have major implications for system-wide throughput. Additional work by van den Berg et al. [16] in reciprocal collision avoidance has informed many graphics-based simulations where pedestrians and vehicles share space under time pressure.

The growing maturity of digital twin technologies has opened new possibilities for urban DRR. Frameworks such as [17] and [18] highlight the role of 3D urban models and sensor feedback in resilience planning. From a participatory planning perspective, Wang et al. [19] integrates GIS data with public engagement tools, though few models achieve centimeter-scale resolution or simulate full-scale evacuations. Notably, graphics platforms like Unity and Unreal Engine have been adapted to support high-fidelity, agent-based urban simulation, as demonstrated in [20] and [21], offering both visual realism and behavioral plausibility.

Our work builds on these foundations by integrating high-resolution spatial data, scenarios with pedestrian and vehicle simulations, and behavioral calibration into a real-time, interactive digital twin of Nicosia's Old Town. This allows for the testing of multi-hazard evacuation scenarios under realistic constraints, such as blocked gates or roads, narrow corridors, etc. providing actionable insights for heritage city DRR planning.

3. Methodology

This section outlines the development and implementation of our simulation framework. We describe the 3D environment setup, network reconstruction, shelter modeling, agent behavior design, and traffic dynamics.

3.1. Simulation environment

All simulations were developed in *Unity*, which offers an integrated 3-D renderer, physics solver, and AI navigation toolkit. The digital twin of Nicosia's Old Town, taken from i-Nicosia, was imported as textured meshes, providing geo-referenced buildings, terrain, and infrastructure elements such as walls and street furniture. This model forms the spatial foundation for evacuation scenario testing.

3.2. Network reconstruction and nav. Layer

The road network was reconstructed using *EasyRoads3D*, manually aligning each segment with the twin geometry to preserve geometric accuracy. The six historic gates were defined as the only legal exit points, reflecting the constraints imposed by the 16th-century fortifications. Due to the military buffer zone in the north, vehicles and pedestrians are restricted from crossing that boundary (as seen in Fig. 1).

To enforce one-way street constraints and prevent unrealistic routing, we used Unity's AI Navigation system to build a detailed *navigation mesh* (nav-mesh), overlaid with directional off-mesh links that encode one-way restrictions. Note that, pedestrian-only paths were excluded from the vehicle navigation layer.

3.3. Shelter modeling

Shelters were represented as dynamic nodes within the simulation. Consultation with Cyprus Civil Defense confirmed two active shelters (with capacities of 180 and 200), and two additional sites pending activation. Each shelter was parameterized with its capacity, and agents were programmed to select the nearest shelter but redirect dynamically if capacity was exceeded. This redirection logic allowed for realistic modeling of queuing, spillover flows, and localized congestion effects.

3.4. Agent modeling

To simulate realistic evacuation behavior, we modeled two main agent types: pedestrians and vehicles. Each type is governed by distinct motion dynamics, perception rules, and decision-making processes. Both agent systems operate in the same navigable environment but respond differently to congestion, routing, and environmental cues.

3.4.1. Pedestrians

Pedestrian agents used Unity's nav-mesh for global path planning and a lightweight motion controller for local steering and skeletal animation. Parameters such as acceleration, avoidance radii, and stopping distance were manually calibrated to mimic real-world walking dynamics. To model varied familiarity with the city layout, and represent heterogeneous knowledge, we implemented a leader–follower mechanism: these agents follow a designated leader until nearing a gate or shelter, then switched to individual pathfinding. This behavior generated emergent platoon dynamics, similar to tourist or group evacuation behavior.

3.4.2. Vehicles

Vehicle behavior posed unique challenges. Unity's default nav-agent was initially adapted for cars, but this yielded unrealistic motion. Instead, we developed a custom *CarController* script. Each vehicle still used the nav-agent path for navigation, but actual motion was computed by applying physical forces and torques to a rigid body, producing more lifelike acceleration, deceleration, and turning. To avoid collisions, we combined Unity's Reciprocal Velocity Obstacles (RVO) module with three front-facing ray-casts (Fig. 2), which detect obstructions and modulate throttle in real time.

Navigation decisions were made using a "fewest-corners" heuristic rather than Euclidean distance, reflecting the complexity of one-way grids. After running a number of simulations, we observed that strictly following these precomputed routes created unrealistic behavior, as real drivers will often detour when they spot heavy traffic. To model real-time driver adaptation, we implemented congestion triggers: each street segment monitored its vehicle density and, if overloaded, increased its nav-mesh cost. Therefore, all streets have their own Navmesh area and a congestion threshold customized to its width and position. Approximately 25% of agents responded by recalculating their route, producing a plausible mix of compliant and opportunistic drivers.

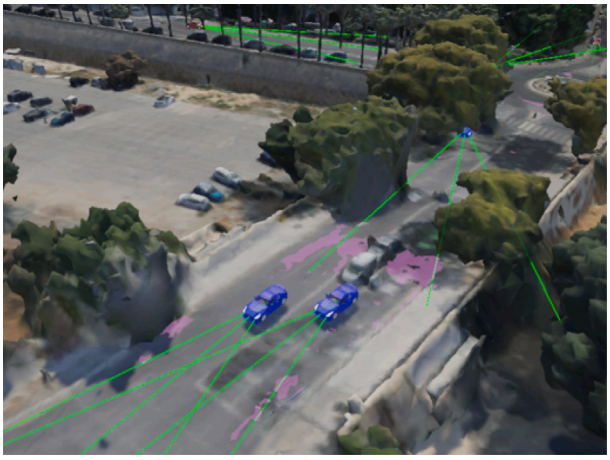


Fig. 2. A snapshot of vehicle simulation within a 3D model of Nicosia. A ray-cast system enables vehicle collision avoidance by detecting frontal obstacles and adjusting throttle accordingly.

3.5. Speed calibration

Pedestrian agents were assigned walking speeds from a normal distribution $\mathcal{N}(\mu = 8 \text{ km/h}, \sigma = 1.5)$, ensuring diversity in crowd dynamics. This distribution was selected based on publicly available data on average adult running speeds over short distances ($\leq$ 600 m), which corresponds to the mean evacuation distance required for agents to reach nearby exits. Vehicle agents were assigned a maximum velocity of 30 km/h, consistent with the speed limit within the Old City boundaries, with modest tail-end overruns permitted on uninterrupted segments to reflect realistic driving variability. These parameter choices were validated through repeated simulation trials to ensure consistency with observed pedestrian and vehicular behaviors.

4. Results

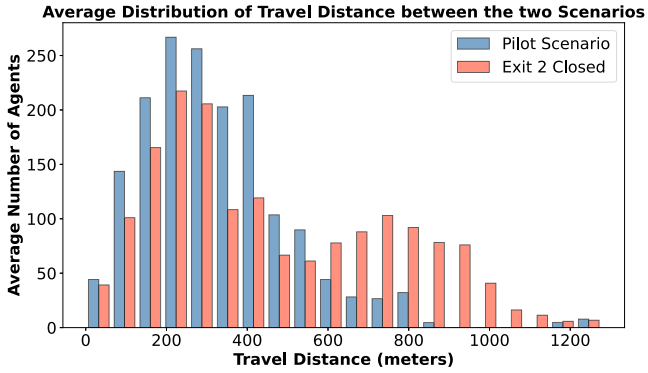
4.1. Implementation details

All experiments were implemented in Unity 6 Preview (6000.0.19f1, DX11). Pedestrian and vehicle routing used the Unity AI Navigation package (v2.0.4), and the road network geometry was generated with EasyRoads3D Free v3 (v3.2.4f3).

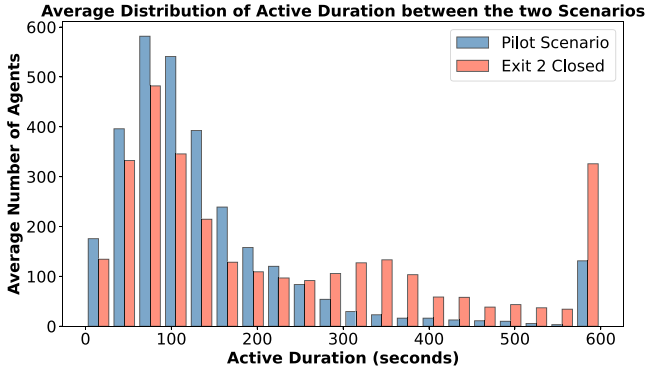
4.2. Experimental design

To examine evacuation vulnerability in Nicosia's walled Old Town under generalized stress conditions, we conducted controlled simulations aligned with the three research questions (RQ1–RQ3). Scenarios were executed with fixed time horizons (10 min for pedestrian-only trials and 15 min for vehicle trials) to enable consistent comparisons. The baseline demand consisted of 3000 pedestrian agents and 700 vehicles, representing peak activity within the city walls. Population levels were informed by the *Census of Population and Housing 2021* [22]. Agents were initialized on the street network and remained stationary until the evacuation signal was issued; the system was closed, with no new entries during each run.

To represent heterogeneity in route knowledge (RQ3), pedestrian scenarios were repeated across five leader-to-follower ratios (80:20, 60:40, 50:50, 40:60, 20:80). Leaders represent informed agents capable of independent navigation, whereas followers rely on nearby leaders for guidance. Metrics reported below are averaged across runs within each scenario suite.



(a) Distribution of Travel Distances Among Agents.



(b) Distribution of Active Duration Among Agents.

Fig. 3. Histogram comparison between pilot and critical exit blockage scenarios regarding travel distance and active duration with average number of agents across five runs.

4.3. RQ1: Evacuation performance under network stress

This section reports simulation outputs describing evacuation clearance, travel distances, congestion patterns, and exit usage under baseline and disruption scenarios. Interpretative implications are presented in Section 5.

Evacuation outcomes were dominated by the availability of a small subset of critical gates. When all exits were available, both pedestrian and vehicle evacuations progressed steadily, but clearance times exhibited long-tailed distributions, indicating that a non-trivial portion of agents required substantially longer travel times to reach safety (Fig. 3).

Pedestrians: With all six gates open (pilot), most agents traveled 100–400 m and cleared within 100–200 s (Figs. 3(a)–3(b)). Leader agents achieved a 95.7% completion rate within the 10-minute window, whereas follower success was substantially lower due to loss of contact under congestion. Exit-use patterns were strongly skewed, with "Dorou Loizou Bridge" serving as the dominant gate.

Disabling "Dorou Loizou Bridge" produced disproportionate delay effects. Distributions broadened in both travel distance and evacuation time (Figs. 3(a)–3(b)), and leader completion decreased to 90%. Rerouted flows concentrated at the "Plateia Eleutherias" exits, which became the primary bottleneck, illustrating how the failure of a single high-demand exit can rapidly saturate alternative corridors.

Vehicles: Under the pilot configuration (all exits open), 80% of vehicles cleared within 335.3 s, and 95% within 462.0 s (Fig. 4). Exit usage was uneven: 76.1% of vehicles used the central/eastern gates (primarily "Archbishop Makariou II" and "Dorou Loizou Bridge"), while only 23.9% exited westward, reflecting the asymmetric structure

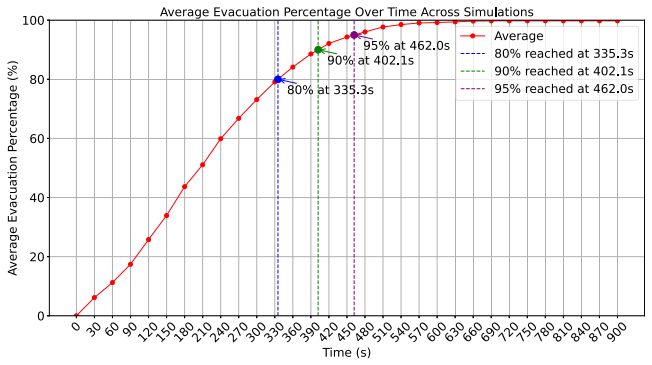


Fig. 4. Cumulative percentage of vehicles evacuated over time in the pilot scenario.

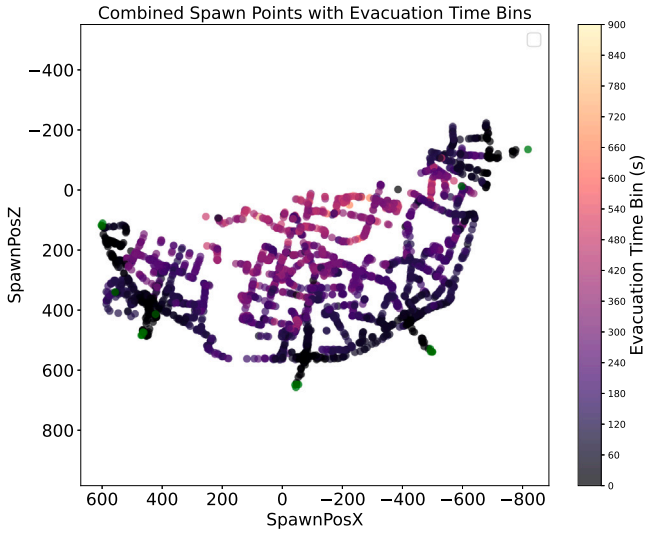


Fig. 5. Vehicle spawn locations color-coded by evacuation time. Darker colors indicate faster clearance. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

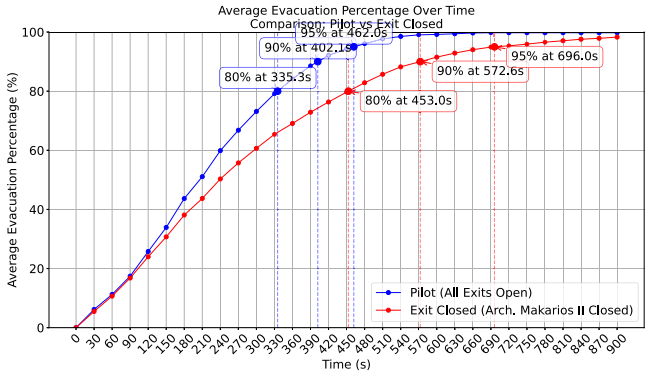
of the street network. Vehicles originating near the center and northern blocks took longer to clear than those closer to the periphery (Fig. 5).

Targeted disruptions confirm this exit-dominance effect. Among the tested blockages, disabling "Archbishop Makariou II" prevented full vehicle clearance within 15 min, demonstrating its role as a system-critical gateway. Fig. fig6a compares cumulative evacuation under the pilot configuration versus closing "Archbishop Makariou II". Fig. 7 shows congestion redistribution under this disruption, with intensified flows near "Dorou Loizou Bridge" and congestion forming around "Vasileos Georgiou B". Further details of all exit closures are provided in Table A.1.

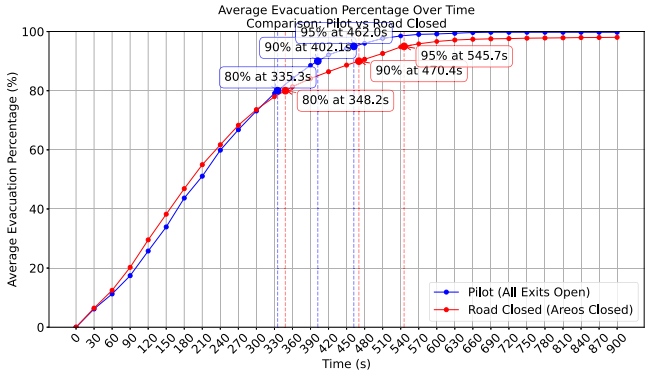
In contrast, several road-segment disruptions produced only modest delays (see Fig. 6(b)), although the closure of the one-way "Areos Street" stranded approximately 1%–2% of vehicles due to one-way constraints. Under moderate stress (e.g., blocking a feeder toward "Dorou Loizou Bridge"), most vehicles successfully rerouted to alternative gates with only limited clearance-time increases, indicating adaptive potential when critical exits remain available. Fig. 8 shows spillover congestion on adjacent streets while the closed segment becomes traffic-free.

4.4. RQ2: Shelter capacity and spatial distribution

To evaluate whether shelter provision improves clearance and reduces exposure, we compared a baseline configuration with two active



(a) Exit Closure.



(b) Road Closure.

Fig. 6. Evacuation Percentage Comparison.

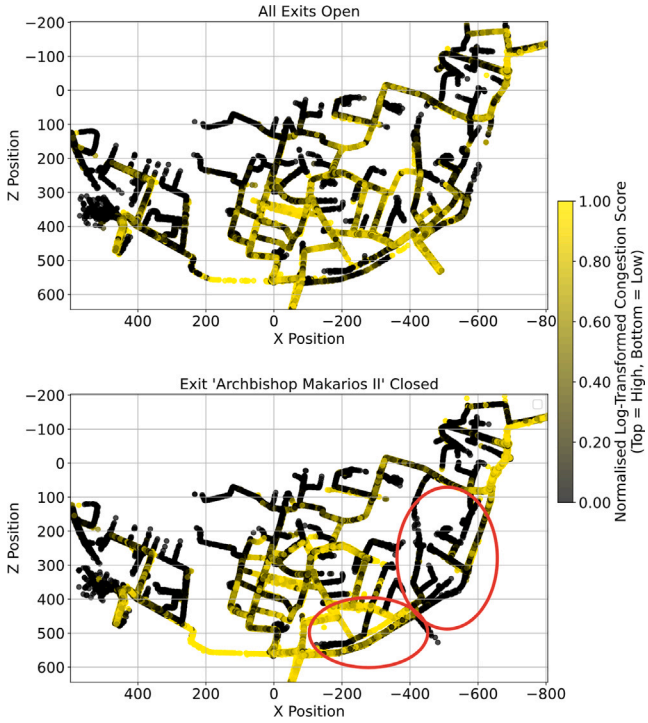


Fig. 7. Congestion Graphs with all exits open (top) VS "Archbishop Makariou II" closed (bottom) where red circles highlight streets that congestion greatly changes. Brighter areas indicate higher traffic density. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

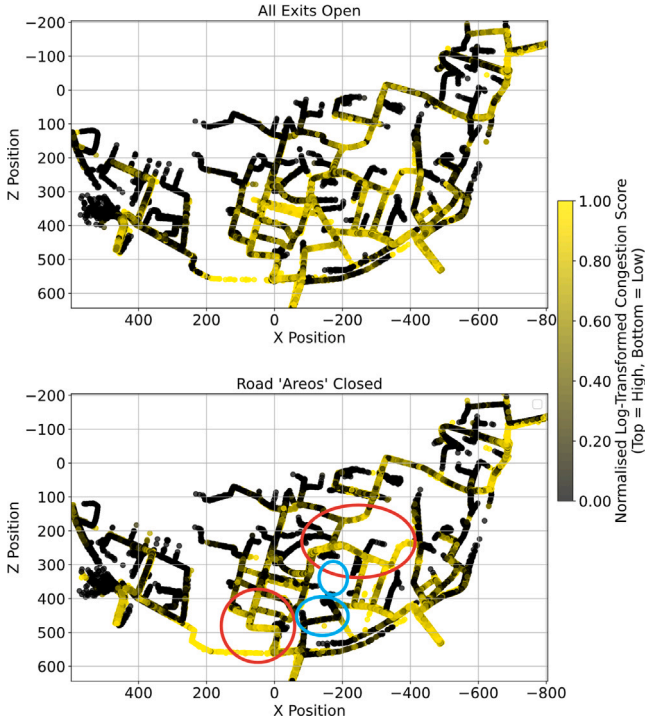
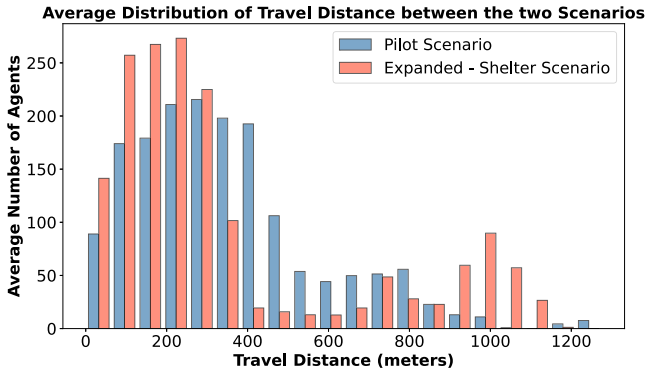
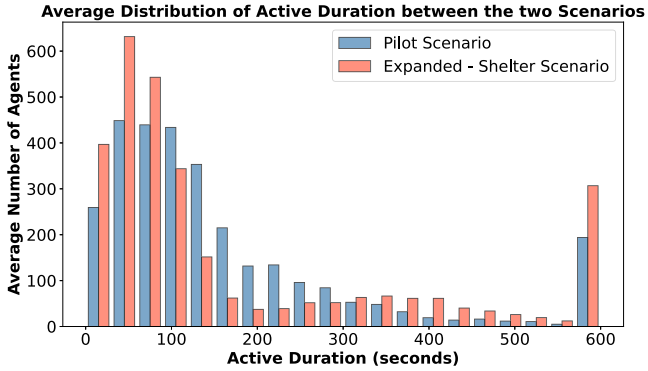


Fig. 8. Congestion Graphs with all exits open (top) VS "Areos Street" closed (bottom). Red circles mark streets where congestion intensifies while the blue circles indicate the now traffic-free closed segment. Brighter areas indicate higher traffic density. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



(a) Travel-distance distribution for pilot (blue) and Expanded-Shelter (red) scenarios.



(b) Evacuation-time distribution for pilot (blue) and Expanded-Shelter (red) scenarios.

Fig. 9. Effect of additional shelters on pedestrian evacuation performance. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

shelters against an expanded configuration with four shelters (two additional candidate sites). Both scenarios used 3000 pedestrian agents and were averaged across the five leader-to-follower ratios.

With two shelters enabled, most agents traveled less than 300 m to reach either an exit or a shelter, and evacuation times peaked below 100 s with a long tail (Figs. 9(a)–9(b)). This pattern indicates that proximity to a safe point is a primary determinant of individual clearance time, while under-served areas generate long-duration outliers.

Activating two additional shelters reduced travel distance for many agents (a new peak below 200 m), but also produced bimodal travel-distance and duration distributions (Fig. 9). A second peak near 1000 m reflects agents who traversed large portions of the network after failing to access a nearby shelter and subsequently redirecting toward another shelter or an exit. These results show that expanding shelter availability can shift demand patterns and introduce additional circulation within the network when capacity and spatial coverage are not matched with effective guidance.

Shelter clearance dynamics: Fig. 10 compares cumulative clearance under the two-shelter (pilot) and four-shelter configurations for leaders and followers. Under the four-shelter configuration (red), the 50% clearance threshold is reached earlier than in the two-shelter pilot (blue), but the pilot reaches higher clearance levels sooner and is the only scenario to exceed 90% within the 10-minute window. Follower agents exhibit the same pattern, with modest early gains offset by mid-run congestion and lower final success rates than in the pilot.

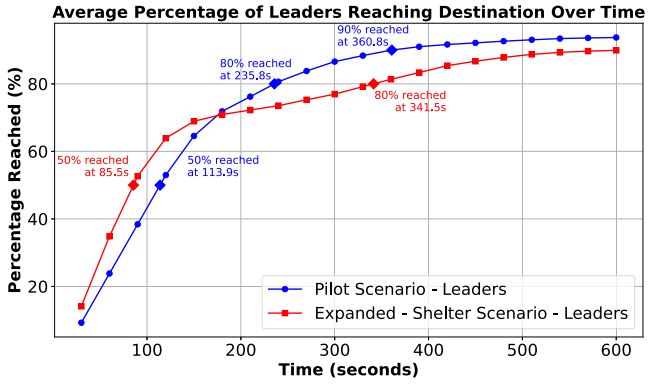
4.5. RQ3: Role of familiarity and guidance

Across all pedestrian scenarios, route knowledge substantially shaped evacuation success. Leader agents consistently achieved higher completion rates and more stable clearance times than followers, including under exit disruption and shelter expansion. As leader share decreased, follower performance degraded sharply, indicating that congestion and disorientation compound when guidance becomes sparse. A tipping point was observed near 70% leader share, below which follower success rates declined markedly. This supports the interpretation of information availability as a capacity multiplier in constrained historic environments.

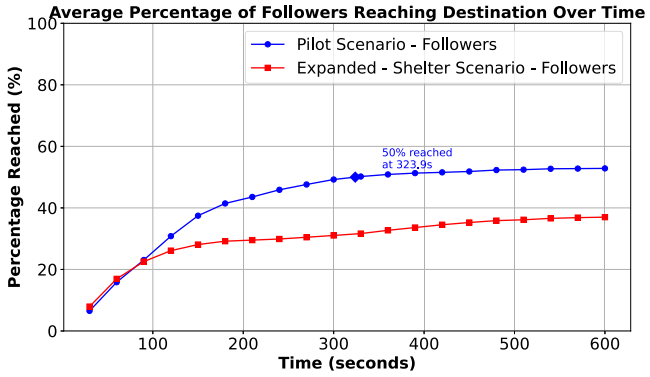
Fig. 11 shows cumulative clearance over time for leaders and followers in the pilot scenario (all gates open) versus the exit-blocked scenario ("Dorou Loizou Bridge" disabled), averaged across runs with varying leader-to-follower ratios. Leaders clear faster in the pilot than in the exit-blocked scenario, and followers exhibit lower success rates with an early plateau under congestion.

5. Discussion

The results highlight structural evacuation sensitivities that are characteristic of walled historic districts: asymmetric street topology, limited exit redundancy, and strong dependence on information flow. Rather than reflecting average performance under ideal conditions, the experiments emphasize how localized failures (e.g., a single gate closure) can trigger disproportionate, system-wide delays, and how shelter provision can unintentionally reshape flows in ways that increase congestion. These effects are also reflected in the long-tailed

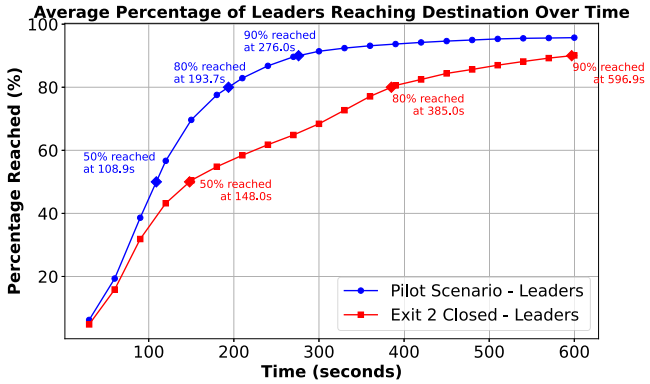


(a) Leaders: cumulative success (2 shelters: blue; 4 shelters: red).

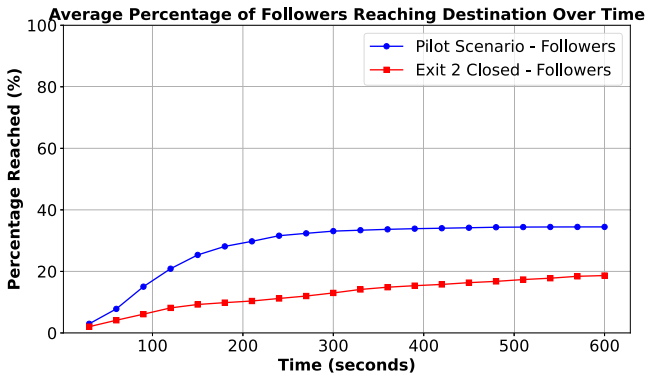


(b) Followers: cumulative success (2 shelters: blue; 4 shelters: red).

Fig. 10. Impact of adding two centrally located shelters on pedestrian clearance.



(a) Leaders: cumulative success over time (Pilot: blue; Exit blocked: red).



(b) Followers: cumulative success over time (Pilot: blue; Exit blocked: red).

Fig. 11. Effect of disabling the "Dorou Loizou Bridge" gate on pedestrian evacuation.

clearance behavior observed across scenarios, where a minority of agents experience substantially longer travel distances and evacuation times than the median population.

5.1. Exit dominance and nonlinear sensitivity

A central finding is that evacuation performance is dominated by a small number of critical gates. Both pedestrian and vehicle experiments show that closing a high-demand exit produces nonlinear degradation: rerouted flows rapidly saturate adjacent corridors, creating secondary bottlenecks that cannot be offset by nominal network connectivity. By contrast, many individual road disruptions yield modest system-level effects unless they interact with one-way constraints that trap agents. From a planning standpoint, this indicates that protecting and maintaining operability of key exits is more consequential than marginal, network-wide upgrades in constrained heritage fabrics.

Consistent with the exit-use concentration observed in the Results section, these findings suggest that limited exit redundancy in walled historic districts creates a small set of operational "single points of failure". In such settings, the practical objective is not only to maximize nominal connectivity, but to preserve functionality at dominant gates and manage diversion demand so that secondary corridors do not saturate under stress.

In historic districts where structural modification is limited, this may involve temporary traffic control measures, reversible street management strategies, or the pre-positioning of emergency personnel at critical gates rather than permanent infrastructure changes.

5.2. Shelters as dynamic network nodes

Shelter expansion altered evacuation dynamics in ways that were not uniformly beneficial. While additional shelters reduced travel distance for nearby agents, the emergence of bimodal clearance distributions indicates that some agents experienced longer routes due to local saturation and subsequent rerouting. This supports a key implication of the study: shelters should be treated as dynamic elements of the evacuation system rather than static endpoints. Capacity increases are most effective when coupled with load balancing and timely guidance that prevents late-stage redirection and unnecessary circulation.

Accordingly, shelter planning in constrained street fabrics should account for how added destinations reshape flow patterns, potentially increasing internal circulation when capacity, accessibility, and guidance are misaligned. Operationally, this points to the value of coordinated shelter assignment (or steering), monitoring of local overload, and contingency options when nearby shelters become unavailable.

5.3. Guidance and familiarity as capacity multipliers

A central finding of this study is that evacuation performance is highly sensitive to population familiarity and guidance. Differences between leader and follower outcomes demonstrate that evacuation capacity is not purely infrastructural. Under congestion and partial network failure, informed agents maintain higher success, while uninformed agents are more likely to stall, disperse inefficiently, or fail to clear within the simulation window.

Taken together with the leader-follower patterns reported in the Results section, the simulations indicate that information availability

can function as an effective capacity multiplier in constrained environments. This effect is particularly important where physical widening, new egress construction, or permanent reconfiguration is limited by heritage constraints.

These findings indicate that information acts as a capacity multiplier, enabling individuals to make adaptive routing decisions that reduce congestion and prevent bottleneck formation. These findings align with the paper's decision-support objective: they motivate investments in pre-event education, clear evacuation protocols, and real-time communication and support systems as complementary measures to physical interventions.

5.4. Relevance for multi-hazard and heritage contexts

Although this study does not model a specific hazard, the stress conditions examined — exit loss, route blockage, and shelter overload — are common consequences of earthquakes, fires, floods, and extreme weather events. The results thus provide hazard-agnostic insights into the structural evacuation vulnerability of historic cities. For heritage districts where preservation constraints limit physical intervention, decision-support tools such as digital twins offer a viable means to explore mitigation strategies before emergencies occur.

5.5. Implications for risk-informed planning in heritage contexts

The asymmetric congestion patterns and rerouting delays uncovered by the model underscore the need for adaptive evacuation strategies. Thus, the study suggests three actionable priorities for DRR in historic walled cities:

1. **Prioritize critical exits:** harden, monitor, and ensure operability of dominant gates under stress.
2. **Enable adaptive flow management:** deploy mechanisms (e.g., dynamic signage, real-time routing directives) that redistribute flows before congestion saturates alternate corridors.
3. **Integrate shelters with guidance:** treat shelter deployment as a coordinated, load-balanced system rather than a standalone capacity expansion.

5.6. Simulation context and planning interpretation

Beyond revealing structural sensitivities, the digital twin functions as a controlled analytical laboratory for stress-testing mitigation strategies before real-world implementation. By enabling systematic manipulation of exits, road segments, shelter configurations, and information conditions, it provides a rigorous platform for evaluating alternative routing policies, load-balancing strategies, and infrastructure protection measures in heritage environments where physical intervention is constrained.

To contextualize these findings, we compared the simulation assumptions with Cyprus's existing disaster management framework, including the Public Warning System (PWS) and the Polyvios Plan, under which evacuation orders are disseminated through SMS alerts, sirens, and local coordination mechanisms. In practice, warning propagation and public response introduce temporal delays and behavioral variability that are not explicitly modeled here.

Our simulations assume idealized informational conditions: agents are already outdoors and respond immediately upon alarm activation. Notably, structural bottlenecks, nonlinear congestion, and clearance delays emerge even under these optimistic assumptions. In real-world evacuations, where communication is staggered, compliance is imperfect, and movement is less coordinated, such vulnerabilities would likely be amplified.

Thus, the digital twin should be interpreted not as a predictive forecasting tool, but as a decision-support and exploratory planning environment. Its value lies in identifying structurally critical exits, congestion-prone corridors, and shelter-induced rerouting cascades, thereby informing risk-informed planning and adaptive management strategies for historic urban areas.

6. Conclusions

This study presented a high-fidelity, hybrid agent-based evacuation analysis of the historic walled Old Town of Nicosia using a geo-referenced 3D digital twin. Rather than predicting evacuation outcomes for a specific hazard event, the objective was to identify structural vulnerabilities and planning sensitivities arising from interactions between historic urban morphology, constrained infrastructure, and human decision-making.

The results demonstrate that evacuation performance in morphologically constrained heritage districts is highly sensitive to exit availability, network redundancy, and the coordination of shelter use. A small number of critical gates dominate system-wide performance, and their disruption produces disproportionate delays and cascading congestion effects. Even under idealized assumptions, where all agents are already outdoors and respond immediately to the evacuation signal, clearance times exhibit nonlinear patterns and long-tailed delays. These findings indicate that structural bottlenecks embedded in legacy urban layouts amplify vulnerability during emergency conditions.

The analysis further shows that capacity-based interventions alone are insufficient. While additional shelters reduce travel distance for some agents, uncoordinated expansion without adaptive guidance can generate rerouting cascades, localized saturation, and increased overall clearance times. Population familiarity and routing guidance function as capacity multipliers: informed agents consistently achieve higher success rates and reduce congestion propagation. Together, these findings highlight that evacuation in historic districts must be approached as a systemic process shaped by infrastructure constraints, spatial configuration, and information flow.

From a disaster risk reduction perspective, three strategic priorities emerge: (i) safeguarding and ensuring the operability of critical exit gates, (ii) integrating real-time routing and communication mechanisms to mitigate emerging bottlenecks, and (iii) aligning shelter planning with dynamic load-balancing strategies rather than static capacity expansion. The digital twin developed in this work provides a robust decision-support platform for testing such interventions prior to real-world implementation and for facilitating stakeholder engagement in resilience planning.

Limitations: Despite the spatial resolution of the digital twin and the explicit modeling of pedestrian–vehicle interactions, several limitations remain. First, agents are assumed to start outdoors and receive evacuation signals simultaneously, omitting building exit delays, staged warnings, and communication bottlenecks that characterize real events. Second, behavioral representation is simplified: agents comply with evacuation intent and do not exhibit panic, hesitation, social attachment (e.g., family groups), mobility impairments, or heterogeneous cognitive responses, all of which can significantly affect evacuation dynamics in historic districts with elderly residents and tourists. Third, hazard evolution is not explicitly modeled. Debris generation, smoke propagation, flood depths, earthquake risk, and time-dependent route degradation are not incorporated, and their inclusion could substantially alter network availability and clearance dynamics. Finally, results reflect the specific morphological and institutional characteristics of Nicosia's Old Town and should not be generalized to other historic cities without contextual adaptation. These limitations position the model as an exploratory and decision-support tool rather than a predictive forecasting system. Its primary contribution lies in revealing structural sensitivities and nonlinear congestion patterns that may not be apparent through schematic planning approaches.

Future Work: We plan to address these limitations by introducing hazard overlays (e.g., debris fields, flood extents, fire spread), dynamic alarm propagation, and heterogeneous agent profiles informed by demographic data. Planned extensions include group-based behaviors, adaptive traffic management strategies, heterogeneous vehicle fleets, and integration with external traffic simulators such as SUMO. These enhancements will enable more realistic multi-hazard scenario analysis while preserving the framework's utility for risk-informed planning in heritage urban environments.

Table A.1
Impact of exit closures on average evacuation delay and network congestion.

Scenario	Avg. addit. evacuation delay*	Main impacts
Archbishop Makarios II Exit Closed	≈ 174 s (≈ 3 min)	Most severe impact. Large load shifts to Doros Loizou Bridge and Vasileos Georgiou B, causing extensive congestion on the peripheral road.
Vasileos Georgiou B Closed	≈ 149 s (≈ 2.5 min)	Second most severe. 56% of vehicles are redirected to Archbishop Makarios, significantly increasing congestion there.
Dorou Loizou Bridge Closed	≈ 146 s (≈ 2.3 min)	Similar impact to Vasileos Georgiou. Archbishop Makarios again absorbs the majority of the traffic flow.
Solomou Square Closed	≈ 53 s (≈ 1 min)	Minor impact. Paphos exit absorbs almost all redirected flow without saturation effects.
Paphos Closed	≈ 11 s	Negligible change. Low initial load, with easy redirection to Solomou Square exit.
Areos Street Closed (street, not exit)	≈ 55 s	Limited increase in time. Alternatives exist in the network, but local congestion appears on cross streets.

*Average difference at the 80%–90%–95% evacuation milestones compared to the pilot trial.

CRediT authorship contribution statement

Marios Stylianou: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **Marios Demetriou:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **Andreas Aristidou:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Impact of exit closures

See Table A.1.

Data availability

Data will be made available on request.

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